deterministic Hamiltonian system taken first, followed by the Ornstein-Uhlenbeck equation for $\boldsymbol{p}$. The flows of each of these systems individually preserves the Gibbs distribution: denoting the Fokker-Planck operators associated to these flows by $\mathcal{L}_H^\dagger$ and $\mathcal{L}_{\mathrm{O}}^\dagger$ we have

$$\mathcal{L}_H^\dagger \rho_\beta = 0, \qquad \mathcal{L}_{\mathrm{O}}^\dagger \rho_\beta = 0,$$

and hence

$$\exp(t\mathcal{L}_H^\dagger)\rho_\beta = \exp(t\mathcal{L}_{\mathrm{O}}^\dagger)\rho_\beta = \rho_\beta.$$

We can consider the propagation of measure resulting from a numerical discretization defined by the composition of these two pieces. For a scheme $[\![\mathrm{HO}]\!]$, we have the distribution first propagated under Hamiltonian dynamics, and then the momenta are evolved as the solution to the Ornstein-Uhlenbeck process, which clearly will preserve the Gibbs measure

$$\exp(h\mathcal{L}_{\mathrm{O}}^\dagger)\left(\exp(h\mathcal{L}_H^\dagger)\rho_\beta\right) = \exp(h\mathcal{L}_{\mathrm{O}}^\dagger)\rho_\beta = \rho_\beta.$$

The problem with this "method" is of course that it requires the exact flow map of the Hamiltonian system. However, this requirement can be relaxed. Given any divergence free vector field $\boldsymbol{f}$, we have

$$-\nabla \cdot (\boldsymbol{f}\rho) = -\boldsymbol{f} \cdot \nabla \rho,$$

thus

$$\mathcal{L}_f^\dagger \rho = -\mathcal{L}_f \rho.$$

If $\mathcal{L}_f$ preserves a first integral $I$, then $\mathcal{L}_f$ preserves any distribution which is a function of $I$. It follows that any energy-preserving, divergence free vector field will always preserve the Gibbs distribution.

All symplectic methods preserve the volume, but for a system with no first integrals except the energy, it is known from a theorem of Ge and Marsden [400] that a symplectic method cannot preserve the system energy exactly (unless the symplectic method is itself a time-reparameterization of the exact solution). The options for volume preserving non-symplectic discretization methods that conserve energy exactly are limited (standard form schemes that rely on calculation of the vector field at a few points cannot achieve this [380]).

A class of methods that do provide the necessary features can be found in the work of Feng Kang [133], referred to as J-splitting by McLachlan and Quispel [261]. Let $\boldsymbol{J}$ be the skew-symmetric canonical symplectic structure matrix. The idea is to consider a splitting of $\boldsymbol{J}$ into a finite number $K$ of skew-symmetric matrices $\boldsymbol{J}_i$, $i = 1, \ldots, K$. This induces a splitting of the Hamiltonian vector field into $K$ vector

fields $\boldsymbol{f}_i = \boldsymbol{J}_i \nabla H$. Each of these vector fields preserves $H$ and, because they are themselves Hamiltonian, their flows are also volume-preserving. Assuming each such vector field is exactly integrable, the system dynamics may be approximated by composing the flow maps for each vector field.

As an example of such a $J$-splitting, we may take $K = \mathrm{N_c}$ with each $\boldsymbol{J}_i$ to be defined by the components of the canonical symplectic structure matrix having to do with the $i$th position and momentum pair, thus the equations of motion for the $i$th vector field become

$$\dot{q}_i = \frac{p_i}{m_i}, \quad \dot{p}_i = -\frac{\partial U}{\partial q_i}(\bar{q}_1, \ldots, \bar{q}_{i-1}, q_i, \bar{q}_{i+1}, \ldots, \bar{q}_{\mathrm{N_c}}),$$

where components with a bar over them are fixed during the calculation. This is, itself, a Hamiltonian system with a single degree of freedom $q_i$ and the potential $\hat{U}_i(q_i)$ obtained by freezing all other components in the original potential function. Any such 1-dof Hamiltonian system is formally integrable (albeit in quadratures). Thus, if we are willing to solve a system using quadrature, we have a numerical method for the Hamiltonian system which preserves volume and energy. We solve each such dynamics from $i = 1$ to $\mathrm{N_c}$, updating the values of each $(q_i, p_i)$ in sequence (the temporally contiguous nature of the updates rules it out as a strategy for parallel computation).

Although this method can be applied to molecular systems, its implementation is challenging and it is not recommended for general computation. Some systems may have a structure amenable to exploitation via this splitting strategy, or it could potentially serve as a starting point in some special purpose methods. For another use of J-splitting in the setting of Metropolis correction of deterministic methods, see [48].

## 7.7 The Evolution of the Distribution Using a Numerical Method

In this section, we shall give an approach that allows the computation of the errors introduced by the discretization of Langevin dynamics for more general potential energy functions. The discretization schemes we shall consider will be those created by composition of the A, B and O splitting pieces described earlier in this chapter, but in principle we can apply this analysis to a wider class of SDE methods. We outline the basic approach for determining the leading order behavior of the discretization error introduced by our SDE integration scheme (see [221–223]), based on expansions due to Talay and Tubaro [357, 358].

In Chap. 3 we employed an asymptotic expansion of the conserved energy-like quantity of a numerical method in powers of the stepsize. By positing such an expansion we are able to calculate the term of any order directly, using a recursive approach. Given such an expansion of the Hamiltonian, we have a corresponding